

Assignment 6

Problem Statement

A mechanical company wants to predict whether a machine will fail or based on parameters: Temperature and Vibration

Dataset

Machine	Temperature	Vibration	Failure
1	High	High	Yes
2	High	Low	Yes
3	Medium	High	Yes
4	Low	Low	No
5	Low	Medium	No
6	Medium	Low	No
7	High	medium	Yes
8	low	High	Yes

A new machine has the following parameters

Temperature = High

Vibration = Medium

$$\text{Entropy} = \sum P_i \log_2 P_i$$

No. of yes = 4

$$P(\text{Yes}) = 4/8 = 1/2$$

No. of No = 4

$$P(\text{No}) = 4/8 = 1/2$$

$$\text{Entropy} = -\frac{4}{8} \log_2 \frac{4}{8} - \frac{4}{8} \log_2 \frac{4}{8}$$

$$= \frac{4}{8} + \frac{4}{8} = 1$$

When Temperature = medium

$$P(\text{Yes}) = 1/2$$

$$\text{no. of yes} = 1$$

$$P(\text{No}) = 1/2$$

$$\text{no. of no} = 1$$

$$\text{Entropy} = -\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right)$$

$$= -\frac{1}{2} \times -1 - \frac{1}{2} \times -1$$

$$= 1$$

When Temperature = low

$$P(\text{Yes}) = 0$$

$$\text{no. of yes} = 0$$

$$P(\text{No}) = 3/4$$

$$\text{no. of no} = 3$$

$$\text{Entropy} = -(0) \log_2 (0) - (1) \log_2 (1)$$

$$= 0 - 1(0)$$

$$= 0$$

Weighted Entropy

$$= \frac{3}{8} \times 0 + \frac{2}{8} \times 1 + \frac{3}{8} \times 0$$

$$= 0 + 0.25 + 0$$

$$= 0.25$$

$$\text{Information gain (Temperature)} = 1 - 0.25 = 0.75$$

For vibration

When vibration = High

no: of yes = 2

$$P(\text{yes}) = 2/3$$

no: of no = 1

$$P(\text{no}) = 1/3$$

$$\text{Entropy} = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3}$$

$$= 0.918$$

When vibration = medium

no: of yes = 1

$$P(\text{yes}) = 1/2$$

no: of no = 1

$$P(\text{no}) = 1/2$$

$$\text{Entropy} = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2}$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

When vibration = low

no: of yes = 1

$$P(\text{yes}) = 1/3$$

no: of no = 2

$$P(\text{no}) = 2/3$$

$$\text{Entropy} = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3}$$

$$= 0.918$$

Weighted Entropy (vibration)

$$= \frac{3}{8} (0.918) + \frac{3}{8} (0.918) + \frac{2}{8} \times 1$$

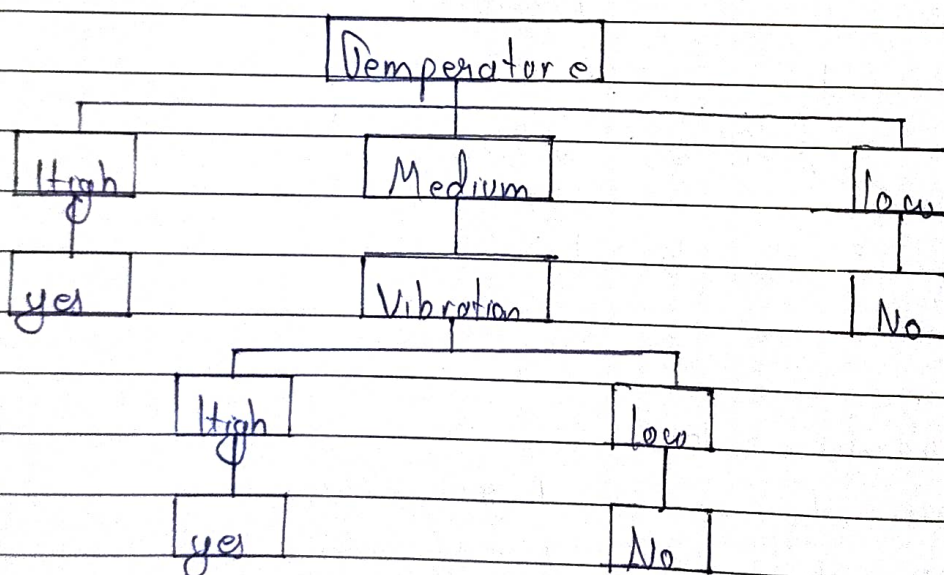
$$= 0.9385$$

$$\begin{aligned}\text{Information Gain (Vibration)} &= \text{Entropy} - \text{Weighted entropy} \\ &= 1 - 0.9385 \\ &= 0.0615\end{aligned}$$

Information gain (Temperature) > Information gain (Vibration)

Root node = Temperature

Decision Tree



EXPLORER

CODING

> New folder

aj.py

Assignment 2_PowerOutput.py

cancer_data.csv

Cancer_predictor.py

Cancer.py

data.csv

Decision_Tree.csv

Decision_Tree.py

KNN_Dataset.csv

KNN_FuelConsumption.py

LoadExtension_ALPHIN.py

LoadExtension.py

machine_failure.csv

Machine_Failure.py

ml_program.py

model.pkl

power_data.csv

sasfids.py

Decision_Tree.py

Decision_Tree.csv

Decision_Tree.py > ...

```
1 import pandas as pd
2 from sklearn.preprocessing import LabelEncoder
3 from sklearn.tree import DecisionTreeClassifier, plot_tree
4 from sklearn import tree
5
6 data = pd.read_csv('Decision_Tree.csv')
7 label_encoder = LabelEncoder()
8 data['Temperature_encoded'] = label_encoder.fit_transform(data['Temperature'])
9 data['Vibration_encoded'] = label_encoder.fit_transform(data['Vibration'])
10 data['Failure_encoded'] = label_encoder.fit_transform(data['Failure'])
11
12 print("Encoded Dataset:")
13 print(data)
14
15 X = data[['Temperature_encoded', 'Vibration_encoded']]
16 y = data['Failure_encoded']
17
18 model = DecisionTreeClassifier(criterion='entropy')
19 model.fit(X, y)
20 input = [[0, 2]]
21 prediction = model.predict(input)
22 print("Prediction Result:", prediction[0])
23 if prediction[0] == 1:
24     print("The machine will fail.")
25 else:
26     print("The machine will not fail.")
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS D:\coding> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/coding/Decision_Tree.py

Encoded Dataset:

	Machine	Temperature	Vibration	Failure	Temperature_encoded	Vibration_encoded	Failure_encoded
0	1	High	High	Yes	0	0	1
1	2	High	Low	Yes	0	1	1
2	3	Medium	High	Yes	2	0	1
3	4	Low	Low	No	1	1	0
4	5	Low	Medium	No	1	2	0
5	6	Medium	Low	No	2	1	0
6	7	High	Medium	Yes	0	2	1
7	8	Low	High	No	1	0	0

C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
warnings.warn(
Prediction Result: 1
The machine will fail.
PS D:\coding>

Python

Python

Python

Ln 16, Col 28

Spaces: 4

UTF-8

CRLF

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Python

Python 3.14 (64-bit)